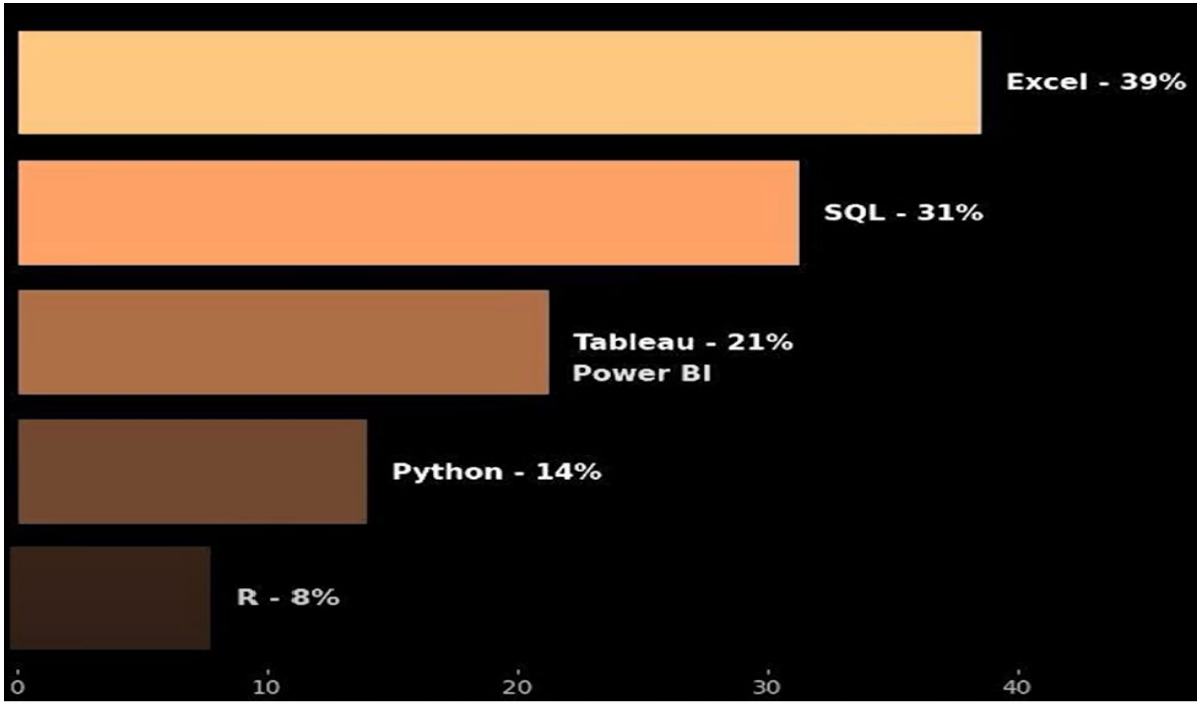
SQL
S1

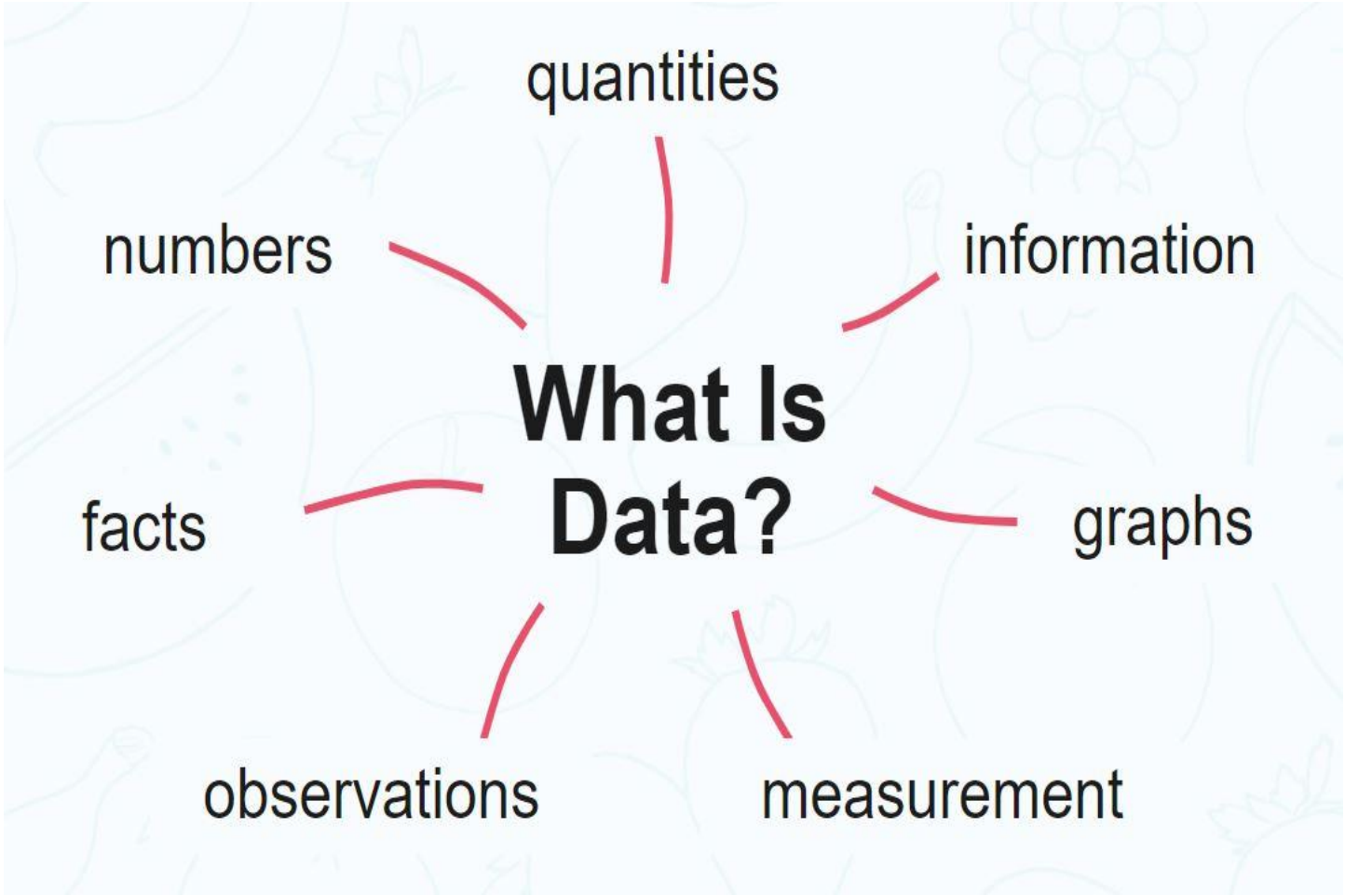
EDC- 2026
Sh. Dowlatshahi

Session 1 : Introduction & program

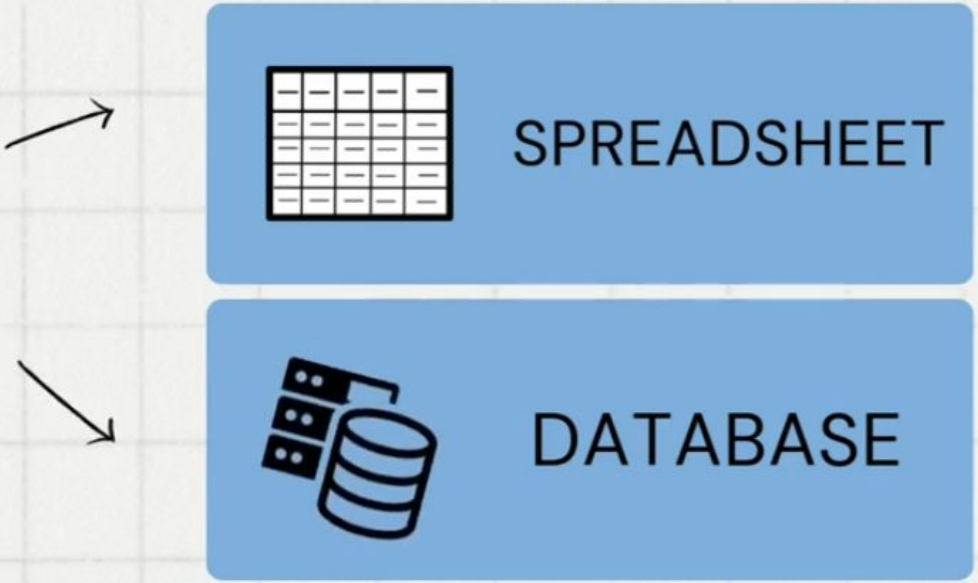
- Ice Breaker which allows you to discover different types of data.
- Why is data important and why analyze it?
- Type of databases, type of software and data professions.
- Download and install PostgreSQL.
- Assignments and grading program for the semester. (S0)

Most in Demand Data Skills in 2025

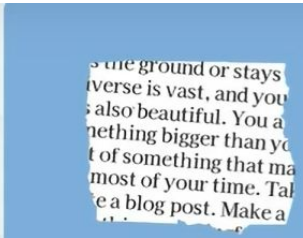



quantities
numbers
information
What Is Data?
facts
graphs
observations
measurement

Structured Data



Unstructured Data



TEXT



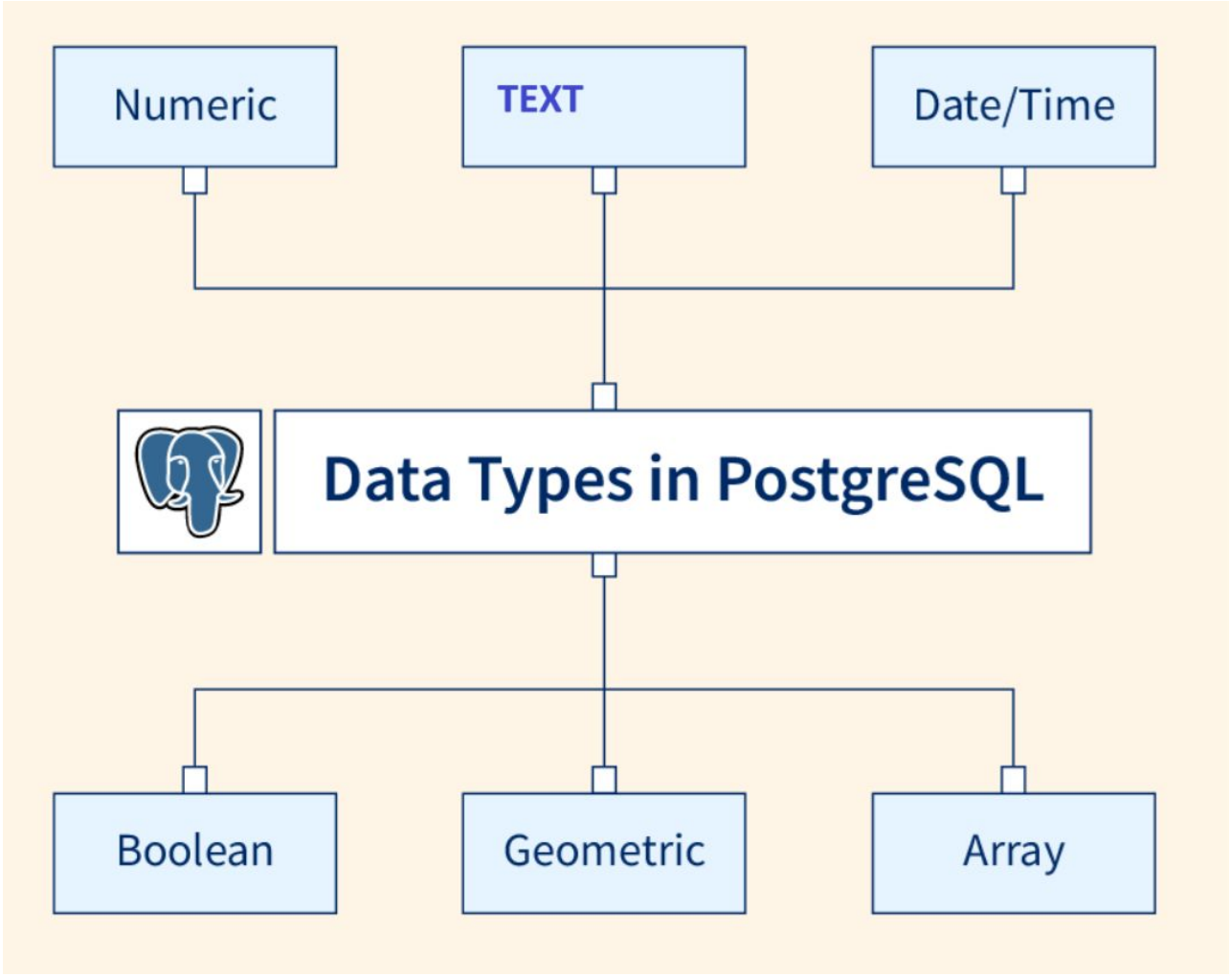
IMAGE



MAP

Activity- PostgreSQL Style




Numeric
TEXT
Date/Time
Data Types in PostgreSQL
Boolean
Geometric
Array

Integer Types

INTEGER

Stores standard integer values

Range: from -2,147,483,648 to 2,147,483,647

The most commonly used integer data type

Integer Types

SMALLINT

Stores small integer values

Range: from -32,768 to 32,767

Integer Types

BIGINT

Stores large integer values

Range: from -9,223,372,036,854,775,808 to 9,223,372,036,854,775,807

Used for very large numerical values

Floating-Point Numbers

REAL

Stores single-precision floating-point numbers

Precision: 6 decimal places

Range: from -3.4E+38 to 3.4E+38

Fixed-Precision Numbers

NUMERIC

Stores numbers with fixed precision and scale

Format: NUMERIC(precision, scale)

Used for financial calculations and other cases where precision is crucial

Floating-Point Numbers

DOUBLE PRECISION

Stores double-precision floating-point numbers

Precision: 15 decimal places

Range: from -1.7E+308 to 1.7E+308

Character Types

CHAR(n)

Stores a fixed number of characters **n**

If the string is shorter than length **n**, it is padded with spaces to the specified length.

If a string exceeding length **n** is inserted, an error will occur.

Character Types

VARCHAR(n)

Stores variable-length strings

Does not pad the string with spaces

Stores only the necessary number of characters

Character Types

TEXT

No length restrictions on the string

Date

DATE

Stores a calendar date (year, month, day)

Format: **YYYY-MM-DD**

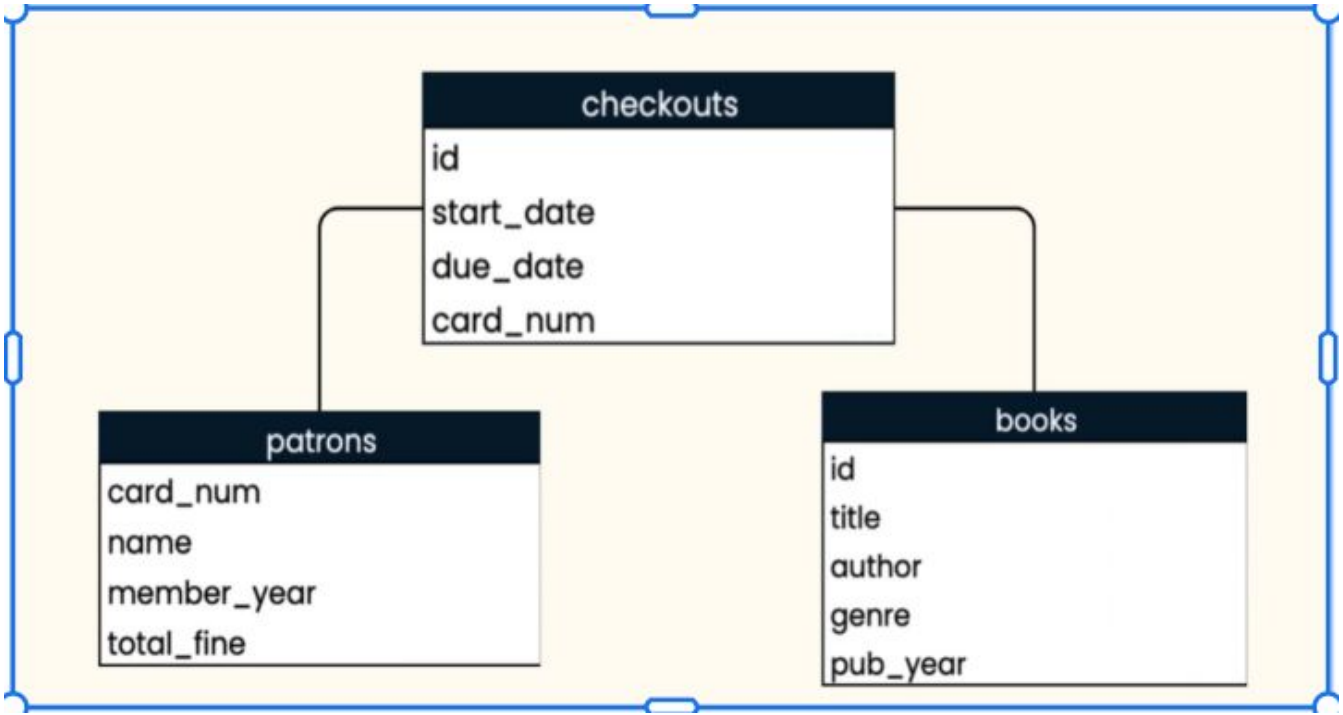
Time

TIME

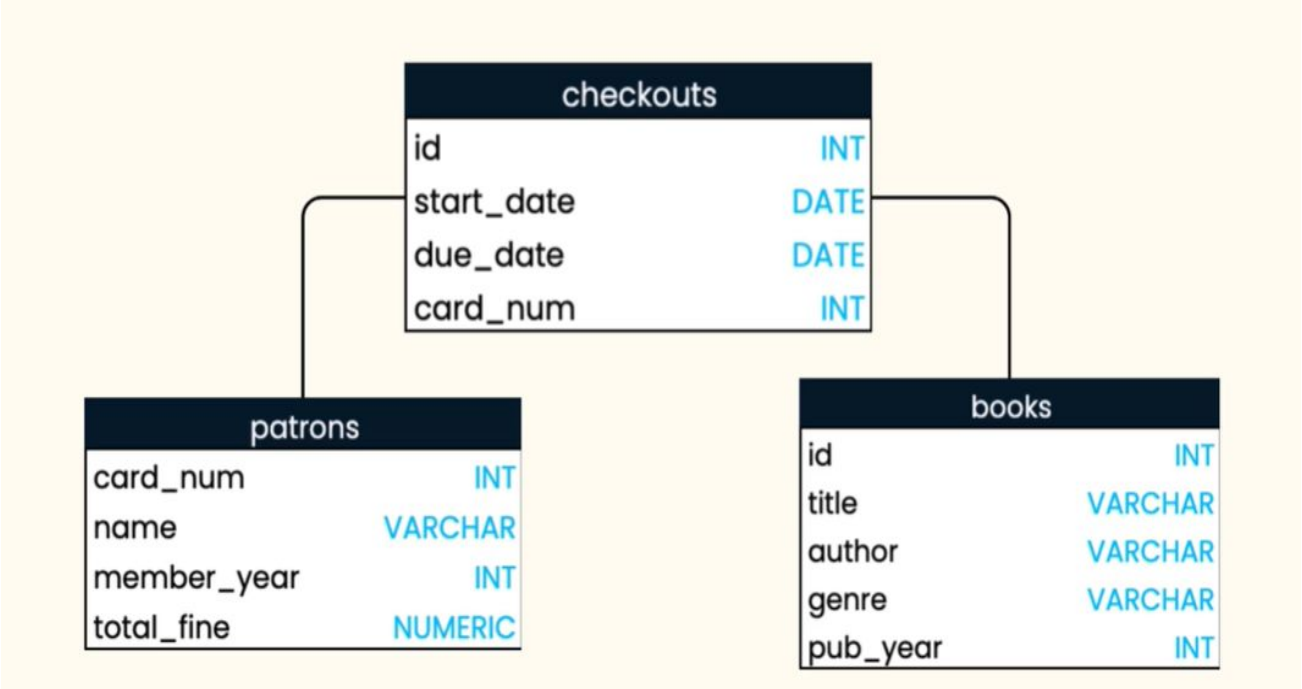
Stores the time of day (hours, minutes, seconds, fractions of a second)

Format: **HH:MI:SS[.MS]**

Example: Identify Data Types

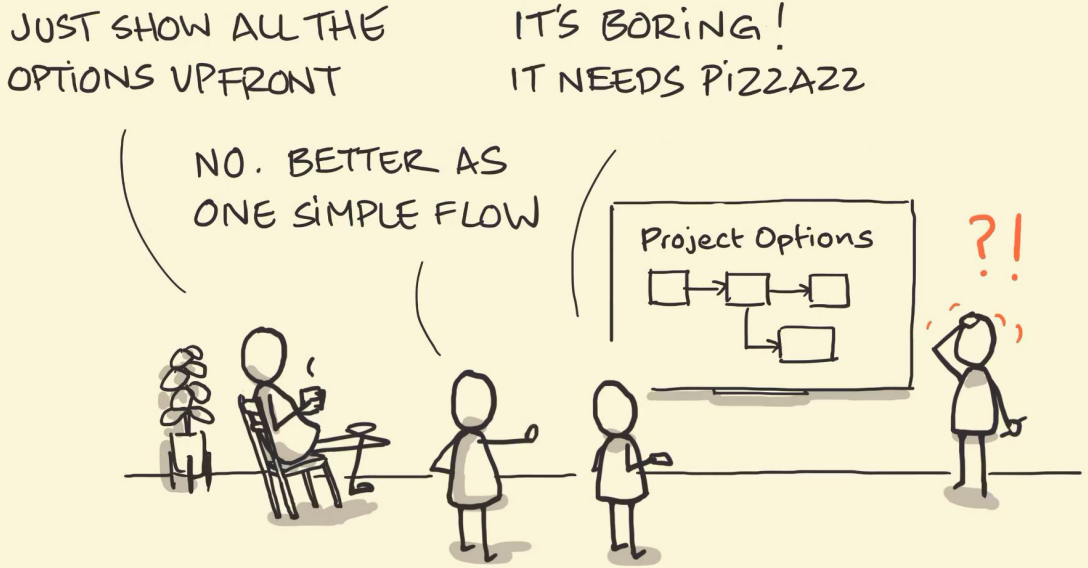


Example: Identify Data Types



Why Is Data Important?

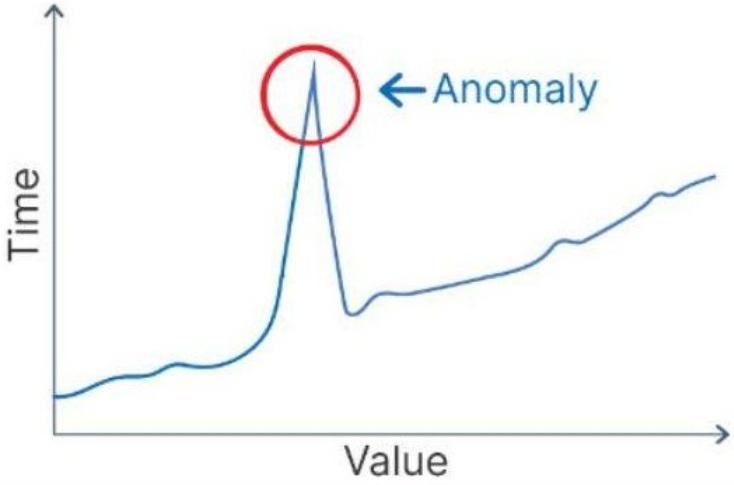




WITHOUT DATA YOU'RE JUST ANOTHER PERSON WITH AN OPINION

— W. EDWARDS DEMING




Anomaly
Time
Value


IMPROVE
PRODUCTIVITY

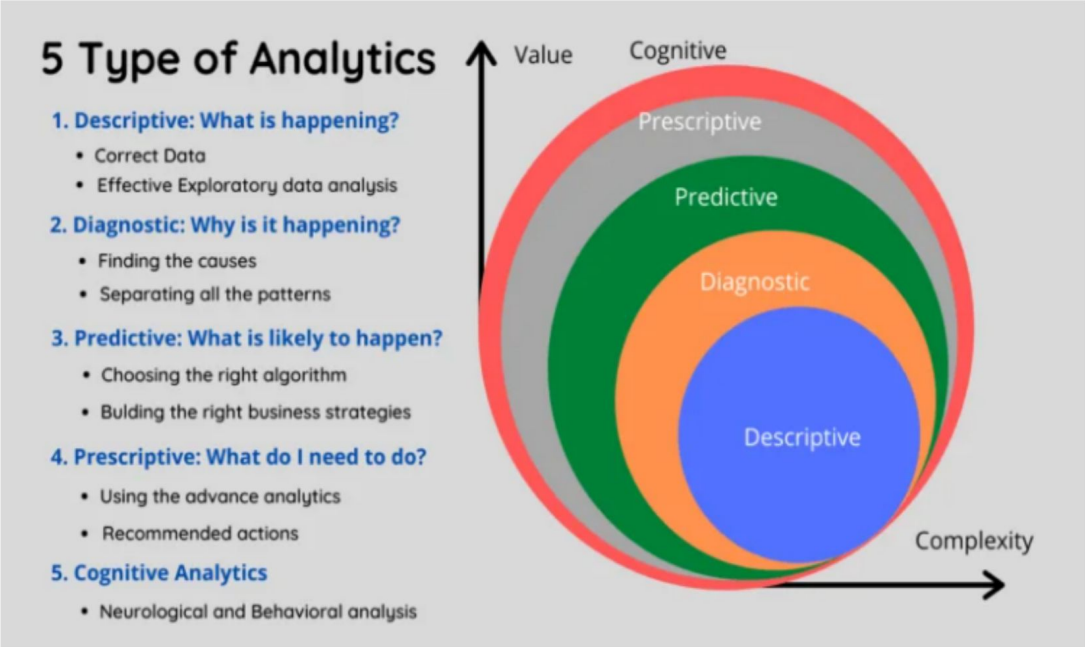

Improve
Efficiency
Macro

Why Do We Analyze Data?




Vision
Mission statement
Objectives
Plan of action
Approach
Tactics


STRATEGY
Quality
Efficiency
Costs


5 Type of Analytics
1. Descriptive: What is happening?
• Correct Data
• Effective Exploratory data analysis
2. Diagnostic: Why is it happening?
• Finding the causes
• Separating all the patterns
3. Predictive: What is likely to happen?
• Choosing the right algorithm
• Bulding the right business strategies
4. Prescriptive: What do I need to do?
• Using the advance analytics
• Recommended actions
5. Cognitive Analytics
• Neurological and Behavioral analysis
Value
Cognitive
Prescriptive
Predictive
Diagnostic
Descriptive
Complexity


books
id
title
pub_year
838
Being Mortal
2015
912
Educated
Non-Fiction
2018
547
Segment of One
Grigsby
Fiction
2022
156
Where the Wild Thing
Sendak
Childrens
1963
patrons
card_num
name
member_year
total_fine
54378
Izzy
2012
9.86
94722
0
45783
2.05
90123
0

What is Table?

books

id	title	author	genre	pub_year
638	Being Mortal	Atul Gawande	Non-Fiction	2015
912	Educated	Tara Westover	Non-Fiction	2018
322	Night	Elie Wiesel	Non-Fiction	1956
156	Where the Wild Things Are	Maurice Sendak	Childrens	1963

What is DataBase (DB)?

patrons

card_num	name	member_year	total_fine
54378	Izzy	2012	9.86
94722	Maham	2020	0
45783	Jasmin	2022	2.05
90123	James	1989	0

books

id	title	author	genre	pub_year
638	Being Mortal	Atul Gawande	Non-Fiction	2015
912	Educated	Tara Westover	Non-Fiction	2018
322	Night	Elie Wiesel	Non-Fiction	1956
156	Where the Wild Things Are	Maurice Sendak	Childrens	1963

checkouts

id	start_date	due_date	card_num	book_id
567	2022-05-13	2022-05-27	54378	638
568	2022-06-10	2022-06-24	54378	322
569	2022-06-27	2022-07-11	45783	156
570	2022-08-14	2022-08-28	90123	912

Database:

A database is an organized collection of data that is structured to allow easy access, management, and updating.



DataBase= SpreadSheet?

Activity

Create a sample database in a spreadsheet with all group members' questionnaire information and save it as a **.csv** file.

Pages in category "Types of databases"

The following 37 pages are in this category, out of 37 total. This list may not reflect recent changes.

A

- Active database
- Animation database

B

- Back-end database

C

- Centralized database
- Cloud database
- *Collection database*
- Collective Tuning Initiative
- Cooperative database
- Current database

D

- Distributed database
- Document-oriented database

E

- EDA database
- Endgame tablebase

G

- Government database

H

- Hybrid transactional/analytical processing

I

- Imieliński–Lipski algebra
- *Incomplete database*

K

- Key–value database

M

- Mobile database

N

- Navigational database
- Non-native speech database

O

- Object database
- Operational database
- Ordered Key-Value Store
- Oxford English Corpus

P

- Parallel database
- Probabilistic database
- Project-Level Aid Database

R

- Relational database

S

- Simple Sloppy Semantic Database
- Suppliers and Parts database

T

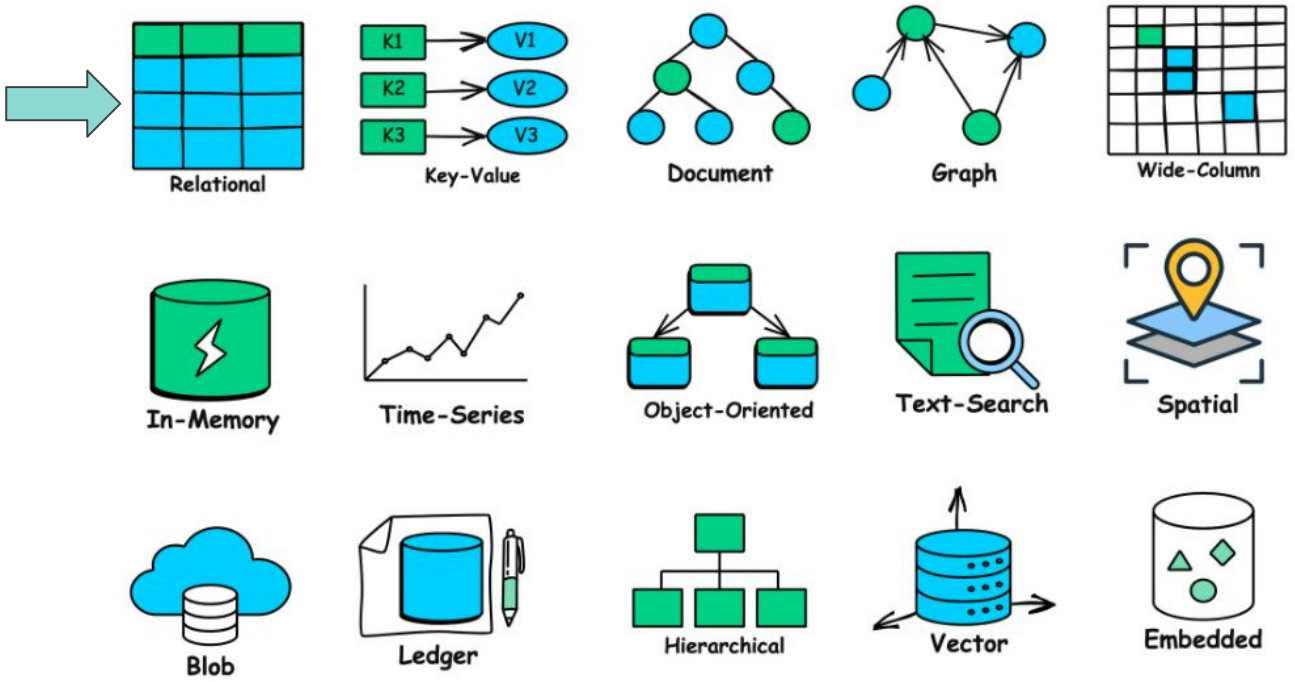
- Triplestore

V

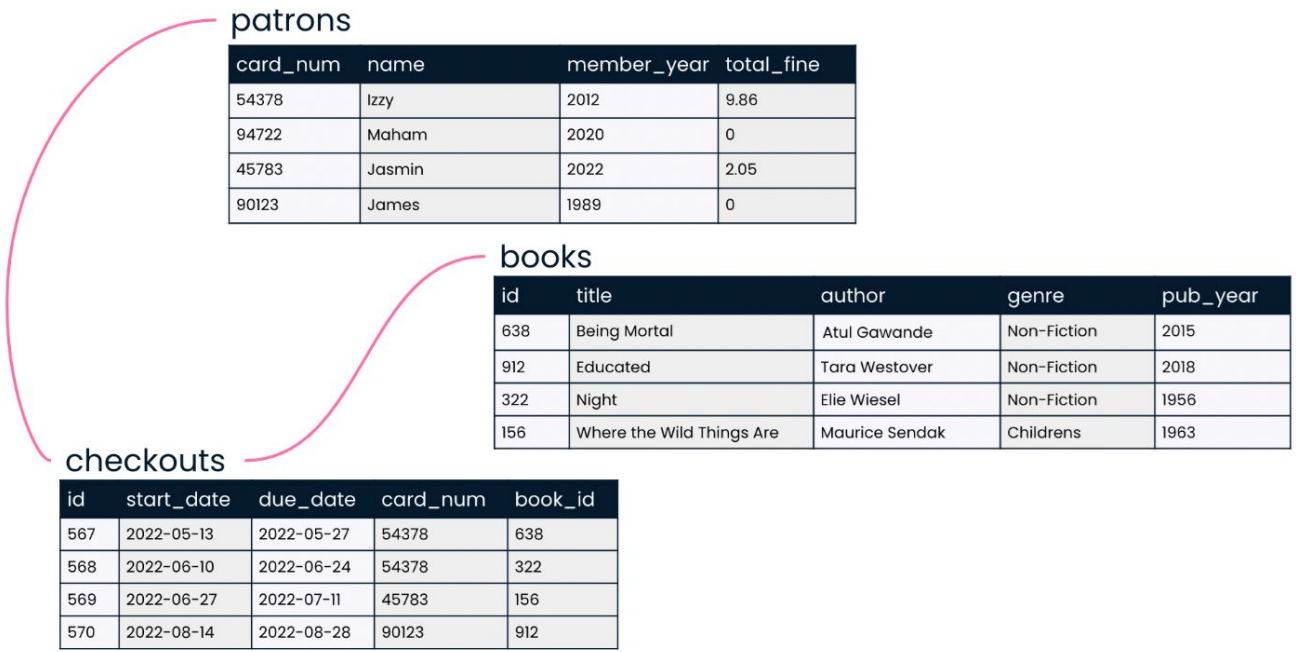
- Vector database
- Very large database
- Virtual private database
- Vulnerability database

X

- XLDB


Relational
K1
K2
K3
V1
V2
V3
Key-Value
Document
Graph
Wide-Column
In-Memory
Time-Series
Object-Oriented
Text-Search
Spatial
Blob
Ledger
Hierarchical
Vector
Embedded

What is Relational DataBase (RDB)?



patrons

card_num	name	member_year	total_fine
54378	Izzy	2012	9.86
94722	Maham	2020	0
45783	Jasmin	2022	2.05
90123	James	1989	0

books

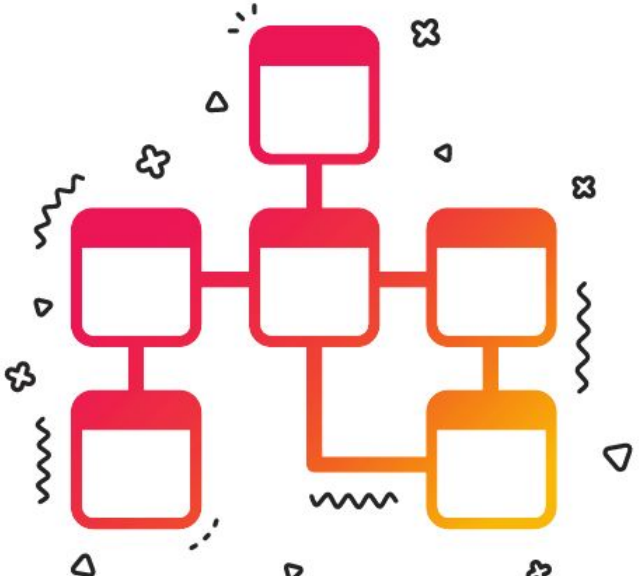
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156	Where the Wild Things Are	Maurice Sendak	Childrens	1963

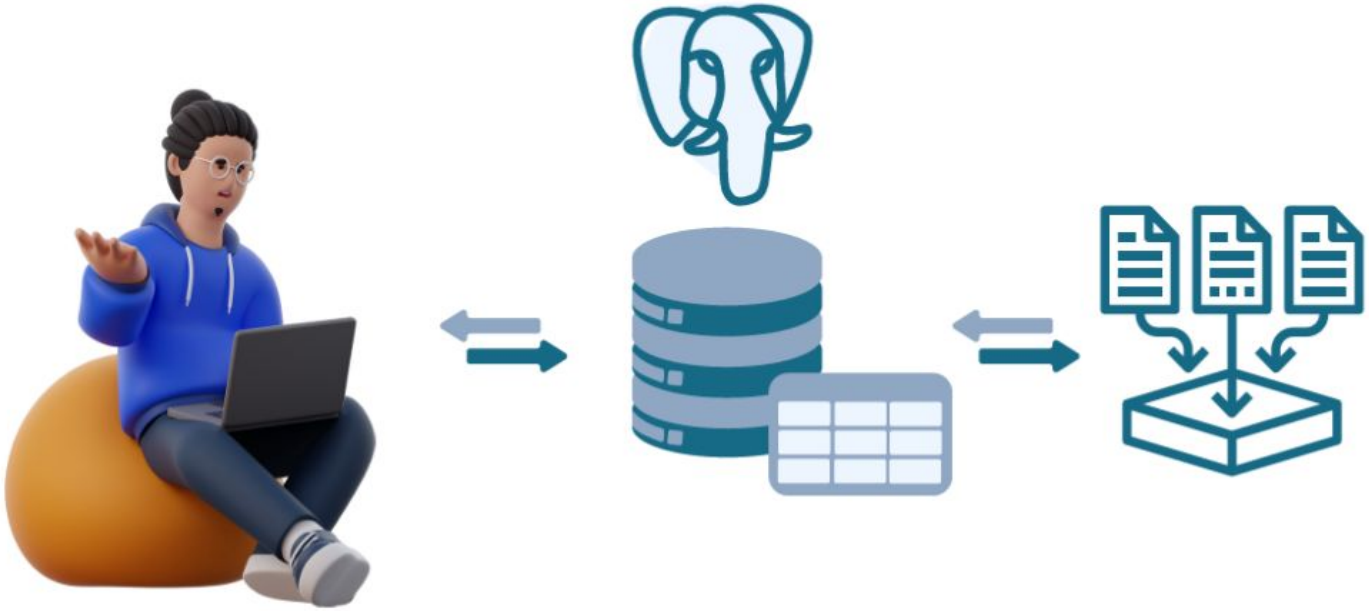
checkouts

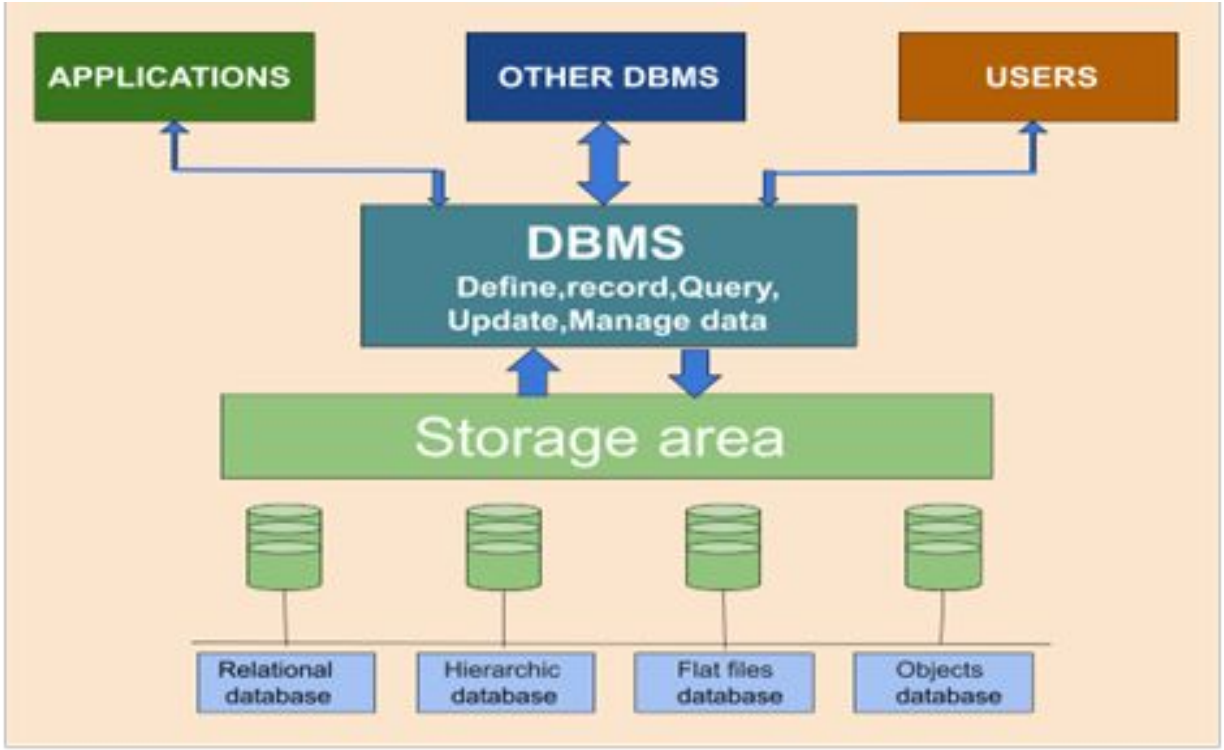
id	start_date	due_date	card_num	book_id
567	2022-05-13	2022-05-27	54378	638
568	2022-06-10	2022-06-24	54378	322
569	2022-06-27	2022-07-11	45783	156
570	2022-08-14	2022-08-28	90123	912

Relational Database:

A relational database is a database based on the relational data model, where data is stored as a set of interconnected tables.

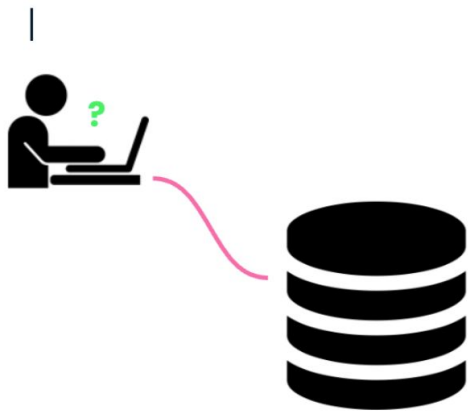




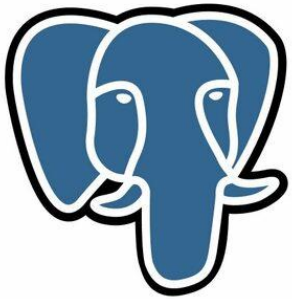

APPLICATIONS
OTHER DBMS
USERS
DBMS
Define,record,Query,
Update,Manage data
Storage area
Relational database
Hierarchic database
Flat files database
Objects database

SQL, or S-Q-L:

```
SELECT *
FROM patrons
LIMIT 30
```









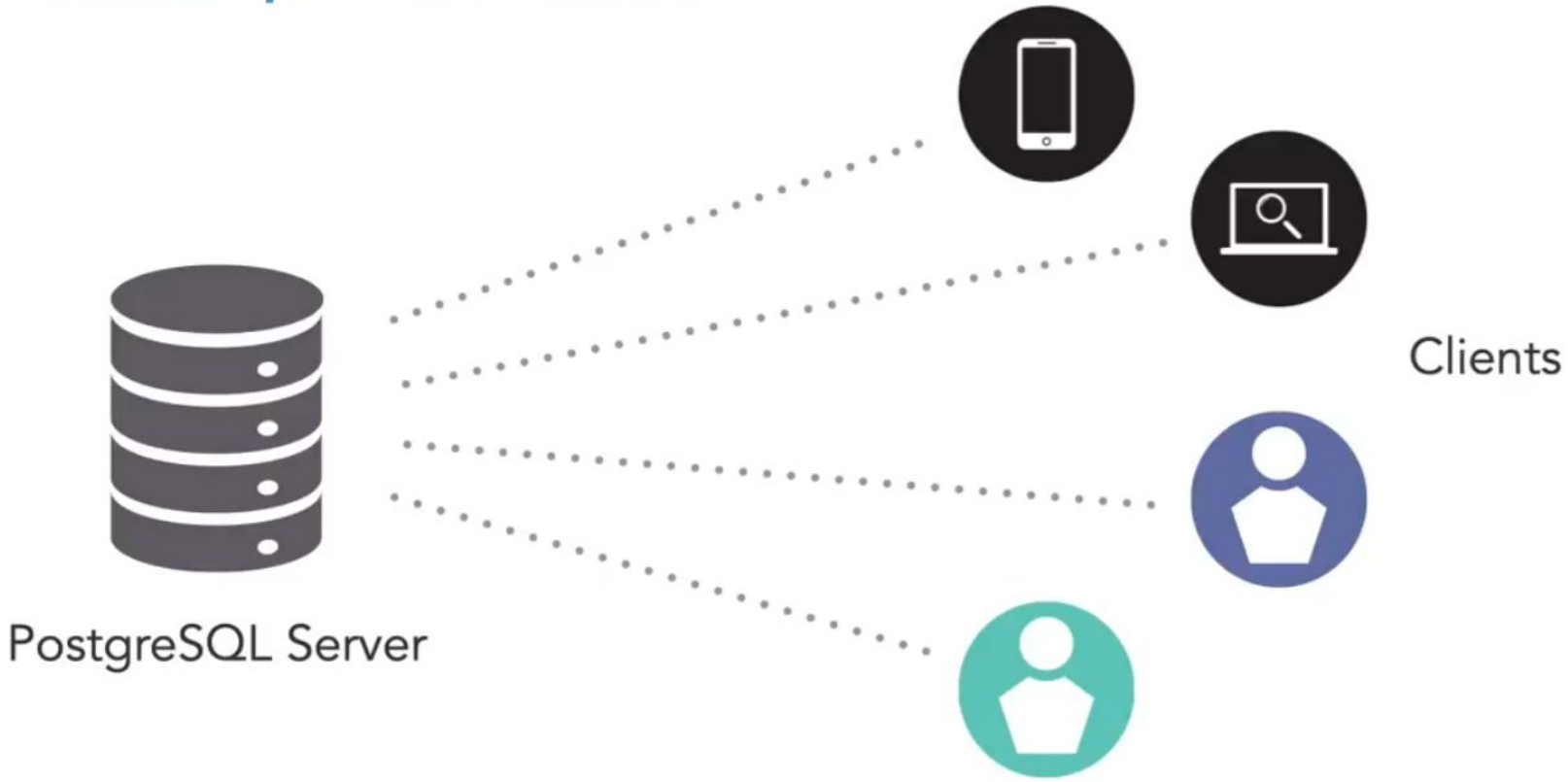


Microsoft®
SQL Server®

ORACLE


PostgreSQL
1
100% Open Source
2
Non-Relational Data Support
3
Best in Class Geographic Information Systems
4
Concurrency Support
5
Extensions
6
Scalability and Performance

Server/Client Model



PostgreSQL

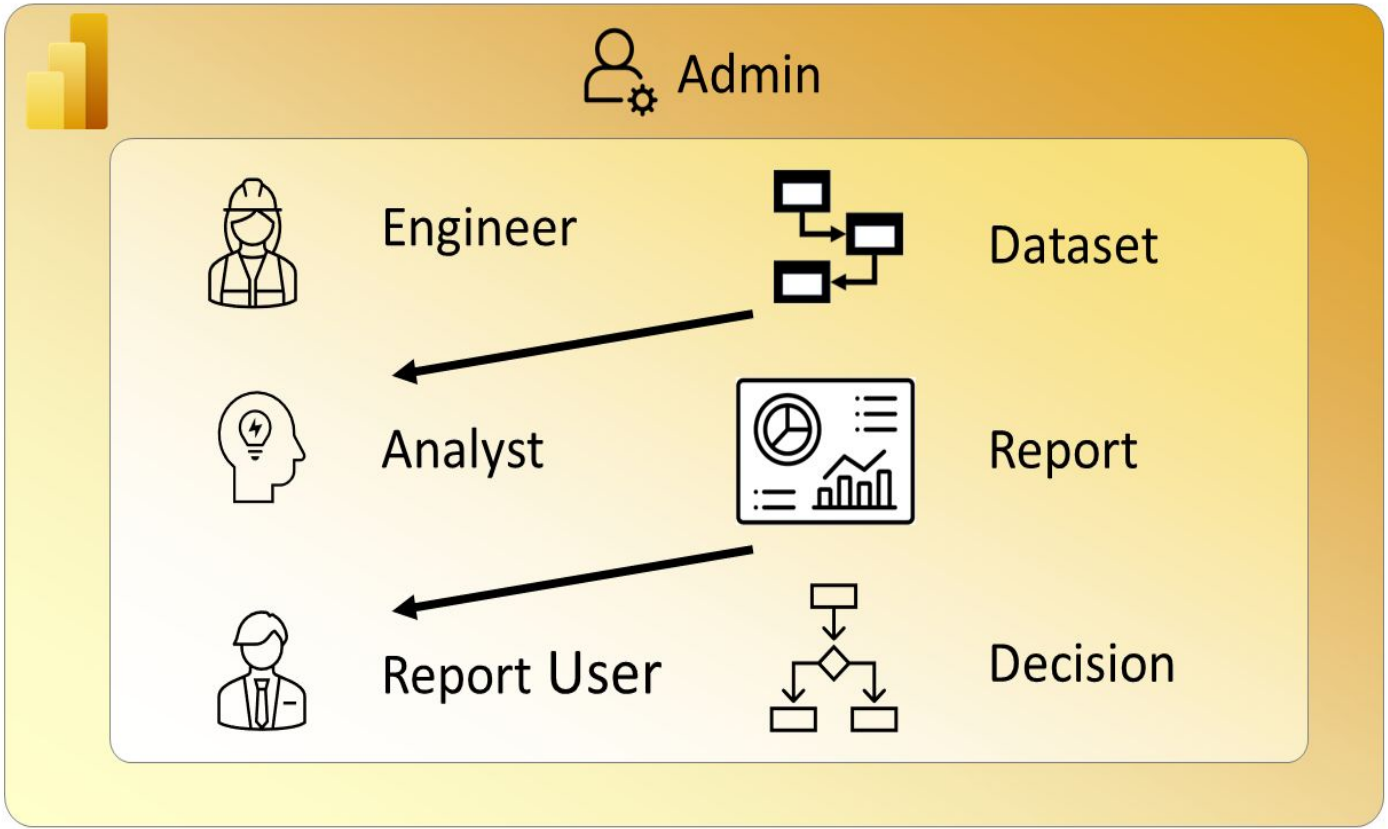
```
SELECT id, name
FROM employees
LIMIT 2;
```

id	name
54378	Darius
94722	Raven

SQL Server

```
SELECT TOP(2) id, name
FROM employees;
```

id	name
54378	Darius
94722	Raven


Admin
Engineer
Dataset
Analyst
Report
Report User
Decision

Data & Software professions



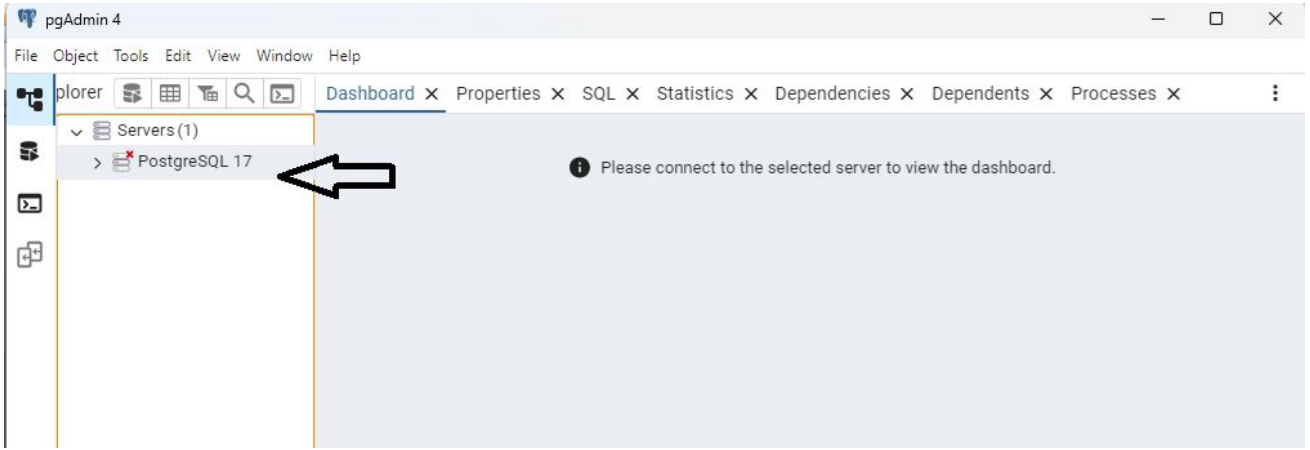
Download & Install

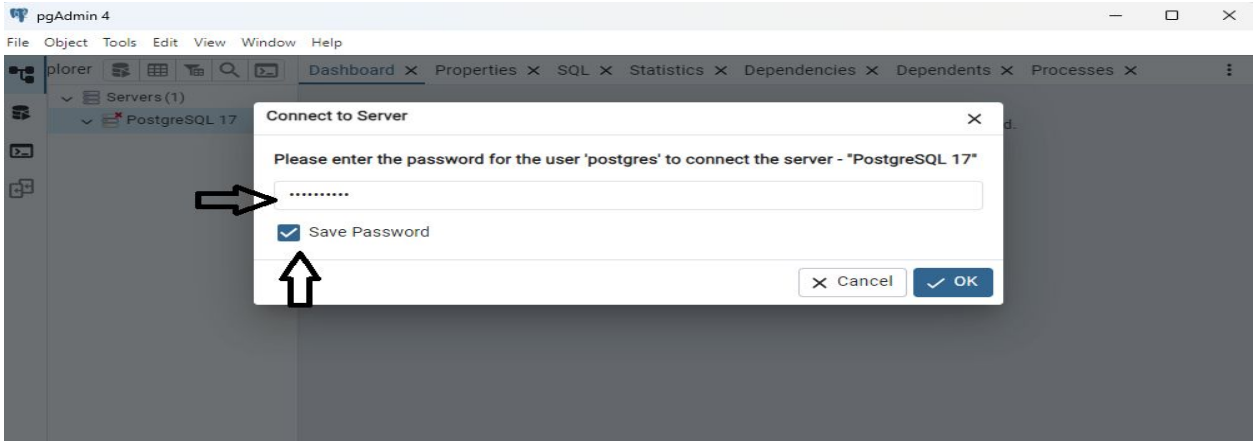
Step1: Download and install postgresql.

postgresql.org

Step 2: Download and install pgadmin.

pgadmin.org


pgAdmin 4
File Object Tools Edit View Window Help
plorer
Dashboard
Properties
SQL
Statistics
Dependencies
Dependents
Processes
Servers (1)
PostgreSQL 17
Please connect to the selected server to view the dashboard.


pgAdmin 4
File Object Tools Edit View Window Help
Dashboard
Properties
SQL
Statistics
Dependencies
Dependents
Processes
Servers (1)
PostgreSQL 17
Connect to Server
Please enter the password for the user 'postgres' to connect the server - "PostgreSQL 17"
Save Password
Cancel
OK